

UGANDA CHRISTIAN UNIVERSITY

A Center of Excellence in the Heart of Africa

DEPARTMENT OF COMPUTING & TECHNOLOGY

Faculty of Engineering, Design & Technology

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1. Display Troubleshooting

The Screen is Flickering

What's likely causing it: Usually, this happens if your monitor's refresh rate is set too low in your Windows settings. It can also be down to a loose, dodgy display cable, or graphics drivers that have become corrupted or outdated.

How to fix it: Head into your display settings and bump up the monitor's refresh rate. Next, give the video cable (whether it's HDMI, DisplayPort, DVI, or VGA) a good push at both ends to make sure it's securely plugged in, or just swap it out for a spare one. Finally, try updating or completely reinstalling your graphics card (GPU) drivers.

The Display Looks Very Dim

What's likely causing it: Someone might have accidentally turned the physical brightness settings right down, or the computer's power-saving modes are throttling the screen's power. In worst-case scenarios, the monitor's internal backlight is on its last legs.

How to fix it: Use the physical buttons on the front or underside of the monitor to crank the brightness back up. If that doesn't work, dive into your PC's power management options and make sure it hasn't turned on an aggressive "eco" or battery-saving mode that dims the display.

Pixels are Dead or Showing the Wrong Colours

What's likely causing it: You've either got a "stuck" sub-pixel that's frozen on a specific colour, or the tiny transistors inside the screen panel are permanently broken.

How to fix it: You can try running a pixel-fixing web app or utility. These work by rapidly flashing different primary colours over the affected area to try and "wake up" the stuck pixel. If it's a true dead pixel (completely black) and won't budge, the screen itself is physically damaged and the monitor will need replacing.

The Screen is Flashing Lines or Weird Coloured Patterns

What's likely causing it: This is a classic sign of a graphics card that isn't sitting properly in its motherboard slot, or a GPU that is seriously overheating because it's choked with dust.

How to fix it: Turn off the PC, unplug it, take the side panel off, and firmly reseat the graphics card back into its PCIe slot until it clicks. While you're in there, grab a tin of compressed air and clean out all the dust from the GPU fans and heatsink. If you're confident doing so, wiping off the old thermal paste and applying a fresh layer to the GPU core works wonders.

The Display Colours are All Wrong

What's likely causing it: Nine times out of ten, a pin is bent or loose inside the video cable or connection port. Alternatively, the monitor's internal colour profile has been messed up.

How to fix it: Unplug the display cable, check the connectors for any bent pins, and plug it back in firmly (or try a brand-new cable). If the cable is fine, use the monitor's physical buttons to open the On-Screen Display (OSD) menu and hit "Reset to Factory Defaults".

The Display is Stuck in Low-Res VGA Mode

What’s likely causing it: Your operating system has lost touch with your actual graphics card, usually because the proper driver is missing or corrupted, forcing Windows to fall back on a generic, basic display driver.
How to fix it: Hop online and download the official driver package directly from NVIDIA, AMD, or Intel, depending on what graphics hardware you have. Installing the proper driver will immediately bring back your monitor's native resolution.

2. CPU and Memory Troubleshooting
The CPU Fan is Making a Horrible Noise
What’s likely causing it: The fan blades are probably clogged up with thick dust, an internal wire is physically dangling into the fan’s path, or the mechanical bearing inside the motor is failing.
How to fix it: Power down the PC and use a soft brush or compressed air to clean the grime off the fan blades. Check around the cooler to ensure no loose cables are catching on the blades as they spin. If the fan still rattles or whines after cleaning, the bearing is shot and you’ll need a new fan.

The Computer Suddenly Reboots, Freezes, or Blurs Out Error Messages
What’s likely causing it: This random instability is usually down to failing RAM sticks, hardware that has worked itself loose, or a CPU that is getting far too hot and shutting down to save itself.
How to fix it: Fire up the built-in Windows Memory Diagnostic tool to scan your RAM for deep-seated faults. If that clears, turn off the PC and physically reseat your RAM modules and graphics card to ensure they have a perfect connection. Keep an eye on your hardware temperatures to rule out overheating.

Upgraded to a Dual-Core CPU, but the PC is Slower and Only Shows One CPU Graph
What’s likely causing it: The operating system is stuck in its old ways literally. It is still configured to use the single-core layout it was originally installed with, or the BIOS configuration hasn't updated.
How to fix it: Restart the PC and tap the delete key to get into the BIOS. Check that the motherboard is actually detecting both cores. If the BIOS sees them but Windows doesn't, the cleanest fix is to do a fresh reinstallation of the operating system so it can reconfigure itself for a multi-core processor.

The CPU Simply Won’t Fit Into the Motherboard Slot
What’s likely causing it: The processor and motherboard socket are completely incompatible, or there are bent pins on the processor or inside the motherboard socket.
How to fix it: Double-check your model numbers. A CPU will only fit into a very specific type of motherboard socket. If they are compatible, check very carefully for bent pins. Remember, you should never have to force a CPU down—it should drop into place smoothly using its own weight.

The Computer Isn't Detecting the New RAM
What’s likely causing it: The RAM sticks aren't pushed in all the way, or the RAM's speed, voltage, or generation isn't supported by your motherboard's chipset.
How to fix it: Take the RAM sticks out completely. Align them properly, and push down firmly at both ends until the plastic clips on the slot snap shut on their own. If they are definitely seated right but still unrecognized, consult your motherboard

manual or check the online Qualified Vendor List (QVL) to make sure that specific type of RAM is actually compatible.

Upgraded Windows and Now the Whole PC Curls Up and Runs Super Slow
What's likely causing it: Old device drivers are clashing with the new Windows update, or your PC simply doesn't have enough physical RAM to handle the heavier demands of the new operating system.
How to fix it: Run Windows Update or visit the manufacturers' websites to install the latest versions of all your drivers. If your drivers are sorted but performance is still poor, check your system specs against the recommended requirements for that version of Windows you might just need to buy a bit more RAM.

3. Power Supply Unit (PSU) Troubleshooting
The PC Restarts, Shuts Off Randomly, or There's a Burning Smell/Smoke
What's likely causing it: This is a critical hardware emergency. You've likely got a severe short circuit, a blown capacitor inside the power supply, or the unit has been completely overloaded.
How to fix it: Pull the power plug out of the wall immediately to prevent a fire. Never open up a power supply yourself to fix it, as the internal components can hold lethal electrical charges. Replace the dead PSU entirely with a brand new one. Make sure you buy a unit with enough wattage to comfortably handle the power needs of your graphics card and processor.

4. Motherboard and Internal Components Troubleshooting
Flashed the BIOS and Now the PC Won't Start at All
What's likely causing it: The update process was interrupted, the file corrupted, or you flashed the wrong version of the BIOS, resulting in a bricked motherboard.
How to fix it: Many modern motherboards feature a built-in automated recovery option (often a specific USB port and button on the back panel). Pop a clean copy of the correct BIOS onto a formatted USB stick and use the manufacturer's recovery utility to re-flash the board.

The Computer Displays the Wrong CPU Info at Boot
What's likely causing it: The motherboard's current BIOS is older than the processor you've installed, meaning it doesn't have the update required to recognize it properly.
How to fix it: Check the motherboard manufacturer's website to ensure your CPU is supported. Boot into the BIOS, load the default factory settings to clear out any old data, and save. If it still reports the wrong info, you'll need to update your BIOS to the latest version.

The Front Hard Drive LED Doesn't Light Up
What's likely causing it: The tiny, fiddly "HDD LED" jumper wire from the computer case is plugged onto the wrong motherboard pins, or it has come loose. (Rarely, the bulb itself has blown).
How to fix it: Grab your motherboard manual and locate the front panel header diagram. Double-check that the "HDD LED" connector is plugged into the exact pins specified, ensuring the positive and negative ends match up correctly.

The Built-in Ethernet (NIC) Has Stopped Working

What’s likely causing it: The network adapter driver has either crashed or become corrupted. If it happened after a storm, a power surge might have fried the physical network chip.
How to fix it: Open Device Manager, locate your network adapter, right-click to uninstall it, and then install a fresh driver downloaded from the motherboard manufacturer's site. If the hardware itself is dead, you can bypass it entirely by installing a cheap discrete PCIe network card or USB-to-Ethernet adapter.

Installed a Brand New Graphics Card and Now There's No Video Output
What’s likely causing it: The card isn't slotted all the way into the PCIe lane, you forgot to connect the power cables, or the card is dead on arrival.
How to fix it: Take the card out and slot it back in firmly, ensuring the retaining clip snaps shut. Double-check that you've plugged in the 6-pin or 8-pin power cables coming from the power supply. If it still doesn't output a picture, try testing the card in a mate's computer to see if the card itself is faulty.

The New Sound Card Isn't Making a Sound
What’s likely causing it: It’s either not sitting properly in its motherboard slot, the drivers aren’t installed, or Windows is still trying to send audio through your old output.
How to fix it: Make sure the sound card is pushed firmly down into its expansion slot. Download and run the official driver installer for that specific sound card model. Finally, click the speaker icon in your Windows taskbar and ensure your default playback device is set to the new card.

5. Storage Devices Troubleshooting
The PC Won't Read an Optical Disc
What’s likely causing it: The disc itself is covered in smudges, dust, or circular scratches, the drive's internal laser lens is dirty, or the drive has been disabled in Windows.
How to fix it: Wipe the underside of the disc with a soft microfiber cloth, moving in straight lines from the centre outward. Try a different disc to see if it’s an issue with that specific piece of media. If nothing reads, check Device Manager to confirm Windows hasn't disabled the optical drive.

The Optical Drive Tray Won't Eject
What’s likely causing it: A mechanical gear inside has jammed, the rubber drive belt has snapped, or a software program is locking the drive closed.
How to fix it: Unbend a standard metal paperclip. Locate the tiny pinhole on the front face of the disc drive, push the paperclip straight inside firmly, and the tray should pop open manually.

The Computer Won't Recognize an External Portable Hard Drive
What’s likely causing it: The USB port you're using isn't providing enough power or is faulty, or your computer's USB controller drivers are playing up.
How to fix it: Unplug the drive and try a different USB port—preferably one directly on the back of the computer attached to the motherboard. If that fails, open Device Manager and update or reinstall your USB Hub controller drivers.

A Memory Card Reader Refuses to Read a Perfectly Good SD Card

What’s likely causing it: The gold contacts on the card are dirty, or the card is formatted in a file system that your operating system doesn't understand.
How to fix it: Grab a clean cloth or a pencil eraser and gently rub the gold contact strips on the back of the card to clear off any grease or dust. If it's clean but still won't open, check if the card was formatted on a device (like a Mac or camera) that uses a file structure your PC can't read.

Transferring Data to or From a USB Flash Drive is Unbelievably Slow
What’s likely causing it: Outdated USB drivers, you've plugged a fast drive into an old, slow USB 2.0 port, or the drive's file structure has become corrupted.
How to fix it: Make sure your motherboard’s chipset and USB controller drivers are fully updated. Ensure you are plugging the drive into a high-speed port (usually colour-coded blue). Finally, right-click the drive in Windows, select Properties, and run the Error Checking tool to sweep for and fix structural file errors.